

Daniel L. Perret

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OVERVIEW

I am conservation biogeographer, forest ecologist, and quantitative scientist with broad training and expertise. My interests fall at the intersection of scientific research, application, and communication, with the overarching goal of improving conservation and management outcomes. While my recent research has focused on the responses of forest systems to global change, I am motivated to creatively apply my broad expertise to a variety of ecological problems. I use many different tools and sources of information in my work, including large complex biodiversity datasets, hierarchical demographic models, aerial and satellite remote sensing data, future climate projections, and cutting-edge ecological theory.

EDUCATION

- PhD, Brown University** 2022
Niche-based perspectives on species' responses to climate change: pines as a multi-scale case study
Department of Ecology, Evolutionary & Organismal Biology
Advisor: Prof. Dov F. Sax
- BS, Stanford University** 2013
Biology (Ecology and Evolutionary Biology, *with Honors*)

PROFESSIONAL EXPERIENCE

- Postdoctoral Research Fellow** 2022 - present
United States Forest Service Pacific Northwest Research Station
Oak Ridge Institute for Science and Education
- I lead quantitative forest ecology research using forest inventories, tree-ring data, field surveys, remote sensing data, and statistical ecological models implemented in R
 - Research achievements in this role have garnered invited commentaries by international scientists in *PNAS* and *Global Change Biology*
 - Organized and led three symposia and workshops at national conferences highlighting innovative research and broadening access to data resources
 - Helped secure >\$200,000 in additional research funding
- Graduate Research Assistant** 2016 - 2022
Department of Ecology, Evolution, & Organismal Biology, Brown University
- Developed and led research initiatives focused on responses of pine species globally to climate change
 - Designed and conducted extensive field sampling campaigns in western US and internationally
 - Published dissertation chapters in *PNAS*, *Ecography*, and *Global Ecology & Biogeography*
- Spatial Analyst, University of Montana** 2014 – 2016
- Used Google Earth Engine, ArcGIS, Python, and eCognition to map spatiotemporal distribution of wetland resources for Greater Sage Grouse across the western US
 - Collaborative interagency project resulted in manager-focused web application
- Field Research Technician, Yellowstone Wolf Project** 2013 – 2014
- Extensive backcountry fieldwork in support of high-profile wolf and mountain lion research
 - Communicated daily with park visitors about wildlife science, conservation, and management

PEER-REVIEWED PUBLICATIONS

- Perret, D. L.,** D. M. Bell, H. S. J. Zald. 2025. Reducing fire severity and extent bolsters subalpine forest resilience to global change through key demographic pathways. *Global Change Biology*.
<https://doi.org/10.1111/gcb.70052>
- Kissling, M. L., M. C. McDevitt, J. Felis, **D. L. Perret**, P. M. Lukacs, *et al.* *In press*. Nesting ecology and movements of an ice-associated seabird, the Kittlitz's murrelet (*Brachyramphus brevirostris*) at the northern edge of its range. *Wildlife Society Bulletin*.
- Perret, D. L.,** M. E. K. Evans, D. F. Sax. 2024. A species' response to spatial climatic variation does not predict its response to climate change. *PNAS*. <https://doi.org/10.1073/pnas.2304404120>
- Perret, D. L.,** D. M. Bell, A. N. Gray, J. D. Shaw, H. S. J. Zald. 2023. Range-wide population assessments for subalpine fir indicate widespread disturbance-driven decline. *Forest Ecology & Management*.
<https://doi.org/10.1016/j.foreco.2023.121128>
- Perret, D. L.,** & D. F. Sax. 2021. Evaluating alternative study designs for optimal sampling of species' climatic niches. *Ecography*. <https://doi.org/10.1111/ecog.06014>
- Rosenblad, K., **D. L. Perret**, D. F. Sax. 2019. Niche syndromes reveal climate-driven extinction threat to island endemic conifers. *Nature Climate Change*. <https://doi.org/10.1038/s41558-019-0530-9>
- Perret, D. L.,** A. B. Leslie, D. F. Sax. 2019. Naturalized distributions show that climatic disequilibrium is structured by niche size in pines (*Pinus* L.). *Global Ecology & Biogeography*.
<https://doi.org/10.1111/geb.12862>
- Donnelly, P. J., B. W. Allred, **D. L. Perret**, N. L. Silverman, J. D. Tack, V. J. Dreitz, & D. E. Naugle. 2019. Seasonal drought in North America's sagebrush biome structures dynamic ecological minimums for sage-grouse. *Ecology & Evolution*. <https://doi.org/10.1002/ece3.4614>

MANUSCRIPTS IN PREPARATION

- Perret, D. L.,** D. M. Bell, H. S. J. Zald. Plots, planes, pixels: multi-scale data integration for dynamic monitoring of forest mortality and risk from insects, diseases, and drought.
- Perret, D. L.,** D. M. Bell, H. S. H. Zald. Dendroecological indicators of sugar pine (*Pinus lambertiana*) vulnerability and decline across its geographic and climatic distribution.
- McLaughlin, B. P., M. Kling, S. N. Miller, **D. L. Perret**, B. J. McGill. An ecology-focused, new-user introduction to the Forest Inventory and Analysis (FIA) Database.

GRANTS AND AWARDS

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| <i>Plots, planes, & pixels: biological disturbance agent estimation. (2023)</i> | \$114,000 |
| PI – Harold Zald, Co-PIs –David Bell, Daniel Perret | |
| ORISE Fellowship supported by interagency agreement between the US Forest Service Pacific Northwest Research Station and the US Department of Energy. | |
| <i>Vulnerability assessment of sugar pine, an iconic species in decline. (2022)</i> | \$100,000 |
| PI – Harold Zald, Co-PIs – David Bell, Andrew Gray, Daniel Perret | |
| ORISE Fellowship supported by interagency agreement between the US Forest Service Pacific Northwest Research Station and the US Department of Energy. | |
| <i>Does ecological niche theory predict tree performance in exotic communities and climates? (2020)</i> | \$9,935 |
| Doctoral Dissertation Enhancement Grant, The Bushnell Graduate Research & Education Fund. | |
| <i>Reading between the rings: the climate future of North American pines. (2019)</i> | \$4,596 |
| Graduate Research, Travel & Training Grant, Institute at Brown for Environment and Society. | |
| <i>Assessing climate-driven extinction risk by measuring performance of North American pines. (2018)</i> | \$4,000 |
| Lewis and Clark Fund for Exploration and Field Research, The American Philosophical Society. | |
| <i>Assessing climate-driven extinction risk by measuring the population performance of pines. (2018)</i> | \$4,977 |

Graduate Research, Travel & Training Grant, Institute at Brown for Environment and Society. <i>Spatial distribution and morphological variation in the mountain beaver (Aplodontia rufa). (2011)</i> Stanford University Undergraduate Advising and Research Small Grant.	\$2,600
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ORGANIZED SYMPOSIA

Demographic perspectives on forest responses to global change. Lead organizer for organized oral session at the US Forest Service Forest Inventory and Analysis Science Symposium. November 2024.

Forest Inventory Data for Global Change Ecology: Opportunities, Challenges, and Innovations. Lead organizer for organized oral session at Ecological Society of America. Portland, OR. August 2023.

An Introduction to Estimating Ecological Change with the US Forest Service Forest Inventory and Analysis (FIA) Database. Lead organizer for workshop at Ecological Society of America. Portland, OR. August 2023.

TALKS AND POSTERS

Plots, planes, and pixels: comparing and integrating datasets to improve forest mortality detection. US Forest Service FIA Science Symposium. November 2024.

Reducing fire severity and extent bolsters subalpine forest resilience to global change through key demographic pathways. US Forest Service FIA Science Symposium. November 2024.

Changing world, changing forests: global, continental, and regional perspectives. Invited seminar, University of Montana. Missoula, MT. April 2024.

Range-wide population assessments for subalpine fir: widespread disturbance-driven decline. USFS Pacific Northwest FIA Operational Meeting. January 2024.

Interactions between climate and disturbance are driving widespread subalpine fir decline. FIA Stakeholder Science Meeting. November 2022.

Interactions between climate and disturbance are driving widespread subalpine fir decline. MtnClimate. Crested Butte, CO. September 2022.

Population-specific responses determine species' vulnerability to climate change: a spatiotemporal case study of ponderosa pine. Ecological Society of America. Montreal, Canada. August 2022.

Tree ring records reveal that climate envelope approaches underestimate population vulnerability to climate change. International Biogeography Society. Vancouver, Canada. January 2022.

Niches, tree-rings, and invasive forests. Brown University EEOB Departmental Seminar. Dec 2020.

Tree rings and niche models as tools for predicting forest responses to climate change. Earth, Itself. Brown University. March 2019. (poster)

Is all niche space created equal? A framework for exploring the structure of the niche. International Biogeography Society. Malaga, Spain. Jan 2019. (poster)

Climatic disequilibrium is structured by niche size. International Biogeography Society, Climate Change Biogeography. Evora, Portugal. March 2018.

Niche size structures climatic disequilibrium in pines (Pinus L.). New York Species Distribution Modeling Symposium. Yale University. December 2017.

TEACHING

Conservation Biology (Biol 1470), teaching assistant Brown University	2020
Guest lectures: <i>Wildlife Management; Wildlife, Climate, and Conservation</i>	
Conservation Biology (Biol 1470), teaching assistant Brown University	2017

PROFESSIONAL WORKSHOPS

Dendrochronology Intensive Summer Course

2018

Laboratory of Tree-Ring Research, University of Arizona

PROFESSIONAL SERVICE

Reviewer for *PNAS*, *Science Advances*, *New Phytologist*, *Ecography*, *Global Ecology & Biogeography*, *Ecosphere*, *Ecology & Evolution*, *Forest Ecology & Management*

REFERENCES

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